CSC186 – Object Oriented Programming
Academic Session October 2025 – February 2026
Lab Assignment 5 - Inheritance

Course Outcomes (CO)	LO1	LO2	LO3
CO1			
CO2	√	√	√
CO3			

1.1 Given the following superclass named `Product` and subclass named `Book`.

Superclass	:	`Product`

Attributes:	:	`String id;      // identification`
		`String title;   // title`

Methods:	:	constructor, mutator, retriver, printer

Subclass	:	`Book`

Attributes:	:	`String author;  // author`
		`int numPages;   // number of pages`

Methods:	:	constructor, mutator, retriever, printer

a) Answer each of the following questions by implementing inheritance concepts:

i. Write the default constructor for both classes
ii. Write the normal constructor for both classes.
iii. Write the mutators/setters for both classes.
iv. Write the accessors/getter/retriever for both classes.
v. Write the printers to return object information for both classes.
vi. Write a method named `calcPrice()` for subclass `Book` to calculate and return the cost of products. The cost for a book is 0.50 cents for each page. The discount rate will be given depends on the cost of the book as shown in the following table:

Cost (RM)	Discount Rate (%)
0 – 200	2
201 – 500	5
>500	10

b) Write an application program that performs the following:

i. Declare an array of objects and input some data into the array of objects.
ii. Display all the object's information, including the cost.
iii. Calculate and display the total cost of the books.
iv. Determine and display the highest and the lowest cost of the book.
v. Count and display the number of books with the minimum of pages is more than 500.

1.2 Sewa4U Online provides rental services for terrace houses and condominium units. The company needs a system to handle its monthly rental collections from tenants. The following class is used to represent a house for rent.

```
public class House {
   private String tenant;  //tenant's name
   private String ICNo;    //tenant's IC number
   private String address; //tenant's address

   public House(…..){ }

   public String getTenant() { return tenant;}
   public String getICNo() { return ICNo;}
   public String getAdd() { return address;}
}
```

a) Write a complete class definition for `House` superclass by considering the following methods.

i. Default and Normal Constructor
ii. Setter
iii. Getter
iv. Printer

b) Define a class `Terrace` subclass to represent a terrace house by using the concept of inheritance. The data members for this class are type (single-storey or double-storey) and corner lot status (Yes/No). You must include the following methods:

i. Default and Normal Constructor
ii. Setter
iii. Getter
iv. Processor
v. Printer

The class should also have a method to calculate and return the rent based on the following table:

Type	Corner Lot	Rent per month
Single-storey	Yes	2000.00
	No	1500.00
Double-storey	Yes	3000.00
	No	2500.00

c) Write a complete Java application to do the following tasks:

i. Declare an array of objects. Read and store all data into the objects.
ii. Count and display the information of tenants for double-storey corner lots.
iii. Display the total rent the company should collect for the month.
iv. Display the type of house rented by a tenant named "Ali bin Abu".

1.3 Given the following `Student` as the super class of `Primary` and `Secondary` classes:

Super Class	:	`Student`

Attributes	:	`String studentID; // student ID`
		`String studentName; // student name`
		`int age; //student age`
		`String nameOfSchool; //school name`
		`//need for extra language, eg: Arabic or`
		`//Mandarin but this is optional for student to`
		`//learn extra language`
		`boolean extraLanguage;`
		`//need for computer course, but this is`
		`//optional for student to learn on extra`
		`//computer software`
		`boolean computerCourse;`

Methods	:	Constructor, mutator, retriever, printer

Sub Class	:	`Primary`

Attributes	:	`// total subject registered for UPSR`
		`int totalSubject;`
		`//false`

Methods	:	Constructor, mutator, retriever, processor, printer

Sub Class	:	`Secondary`

Attributes	:	`int level; //three levels being offered`
		`// 1 -> PMR, 2 -> SPM, 3 -> STPM`

Methods	:	Constructor, mutator, retriever, processor, printer

Sinar Tuition Centre is offering new promotion for UPSR, PMR, SPM and STPM students who are going to sit an important examination this year. The promotion includes language studies and computer courses. For language studies, students can only choose one language from the package. Whereas, for the computer course, the student will be exposed to basic software such as Window 10 and Microsoft Office.

Students are required to take the subject for the appropriate examinations, whereas, they have options either to enroll in the language studies or computer course. The details regarding the subject, language studies and computer course fees are shown in Table 3.1 and 3.2 respectively.

Examination Type	Subject Fees (RM)
UPSR	50.00
PMR	150.00
SPM	200.00
STPM	250.00

Table 3.1 Details of package fees for UPSR, PMR, SPM and STPM

Course Type	Course Fees (RM)
Language	100.00
Computer	150.00

Table 3.2 Details of language and computer course fees

a) Write a complete class definition for superclass and subclass by considering the following methods:

i. The Default and Normal Constructor methods.
ii. The mutator methods for each attribute.
iii. The accessor methods for each attribute
iv. Write the printer method to return the object's information.
v. Write the processor method named `calculatePrimaryFees()` which calculates and returns the total tuition fees for subclass of `Primary`.
vi. Write the processor method named `calculateSecondaryFees()` which calculates and returns the total tuition fees for subclass of `Secondary`.

b) Write a Java application which uses the concept of inheritance to:

i. Store data into an array of objects. The number of data to be stored and information on each student is given by the user.
ii. Calculate and display the total fees collected from students for each examination type .
iii. Determine total number of `Primary` students who have registered for both computer and extra language
iv. List the student id and name for those who have registered **FIVE (5)** subjects for UPSR.

1.4 Hotel HIG Sdn. Bhd offers varieties of services to their customers. The price imposed depends on either they are registered as a member or non-member. Given the following superclass named Customer and subclasses named Member and NonMember

Superclass	:	`Customer`
Attributes	:	`String custName;   //customer name`
		`int icNo;          //identification number`
		`String address;    //address`
		`String roomType;   //room type`
		`int bookingNo;     //booking number`
		`int day;           //number of days`
Methods	:	Constructor, mutator, retriever, printer

Subclass	:	`Member`

Attributes	:	`int memberNo;          //member number`
		 `String dateExpired;    //date expired`

Methods	:	Constructor, mutator, retriever, printer

Subclass	:	`NonMember`

Attributes	:	`//either booking for spa treatment or not`
		 `boolean spa;`
		 `//number of treatment for spa`
		 `int noTreatment;`
		 `//either booking for breakfast or not`
		 `boolean breakfast;`

Methods	:	Constructor, mutator, retriever, printer

a) Answer each of the following questions by implementing inheritance concepts:

i. Write the default and normal constructor for superclass and both subclasses.
ii. Write the mutator for all classes.
iii. Write the retriever method for each attribute for all classes
iv. Write the printer method for all classes.
v. Write a method named `calculatePayment()` for each subclass to calculate and return the total price made by each customer. Details of the charge for room per day are shown in the following table:

Room's Type	Price(RM)/Day
Standard	180.00
Superior	220.00
Deluxe	280.00
Executive	400.00

Additional charges will be imposed to non-member customers if they want to include breakfast and spa services. Meanwhile, there is no extra cost for members. Details of the charges as shown in the following table:

Item	Price (RM)
Breakfast	30.00/day
Spa	150.00/treatment

b) Write an application class called `CustomerApp` to perform the following:

i. Declare and create an array of object customer objects. Then, input some data and store the data onto the object.
ii. Calculate and display the nett price of all customers.
iii. Calculate and display the grand total price of all customers.
iv. Determine and display the customer information with the highest payment.
v. Determine and display the total number of non-member customers who have breakfast and spa services.